

System Design

examples

customers

PIZZA

1 chef

+ 1 backup chef

optimize process and increase throughput using the same resource.

→ Vertical Scaling

prepare beforehand at non peak hours

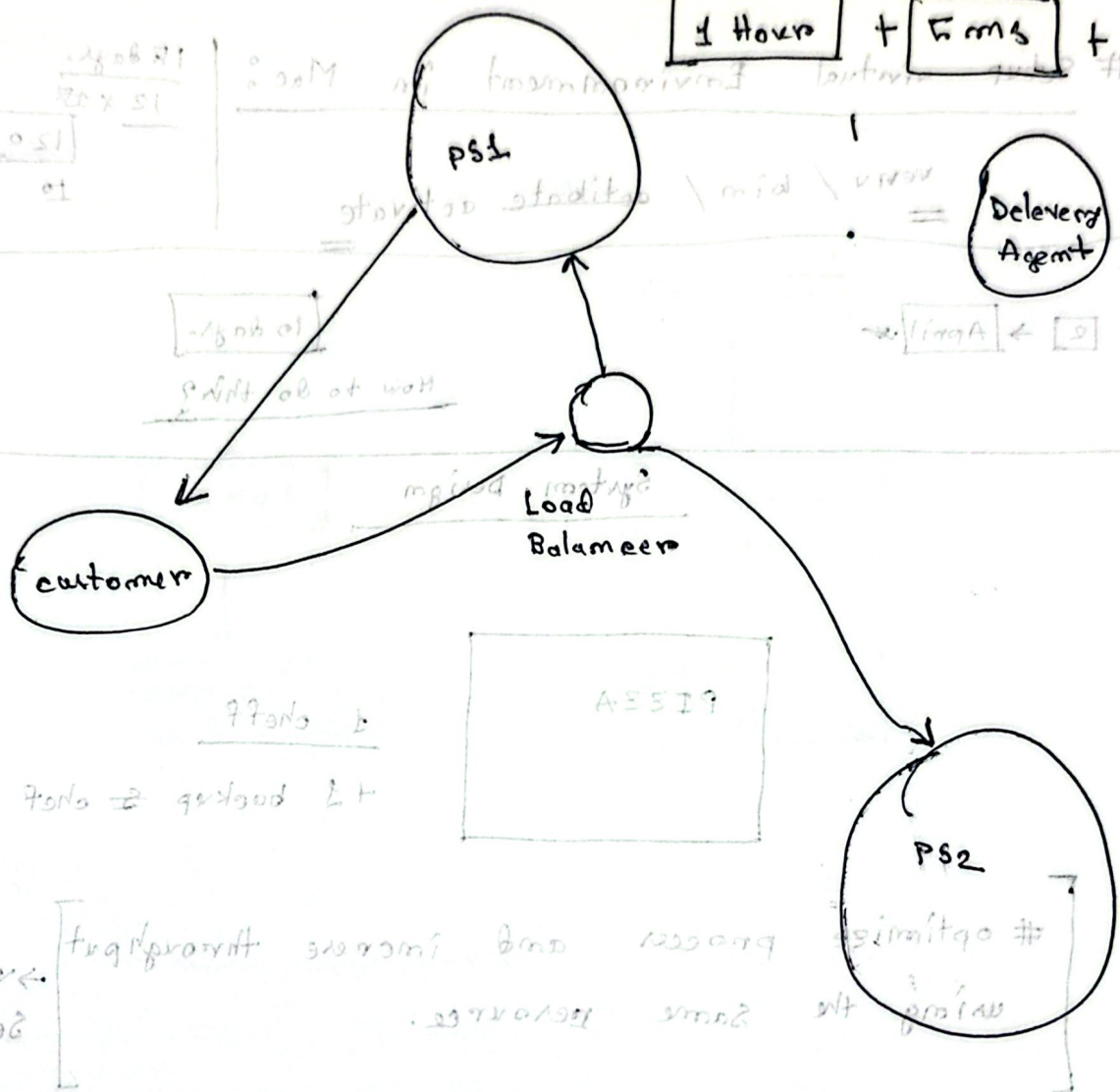
keep backup and avoid single point failure.

[# Hire more resources] - Horizontal Scaling.

- Microservices Architecture.

- Distributed Systems.

- Decoupling



Metrics and logging

Extensible.

High level Design VS low level Design

High ALD

- Deploying on Servers.

- How the two systems will interact with each other.

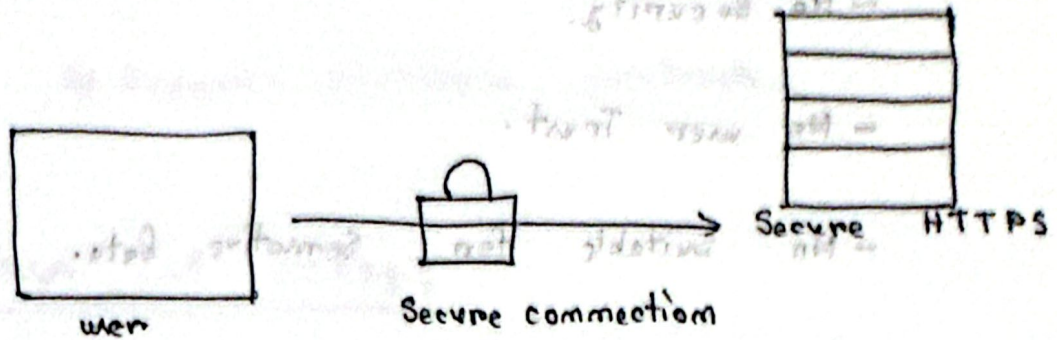
LLD:

- Design class.

What is HTTP and HTTPS protocol?

HTTP = Hypertext Transfer protocol

HTTPS = Hypertext Transfer protocol Secure



HTTP:

operates on port 80 by default

- It transfers data in plain text. meaning content is not protected.

- Because it is unencrypted, attackers can modify the data easily.

- used for not sensitive websites where security is not concerned. [like public blog, website info pages]

Advantages:

Simplicity: Easy to implement and use since it does not require complex encryption mechanisms.

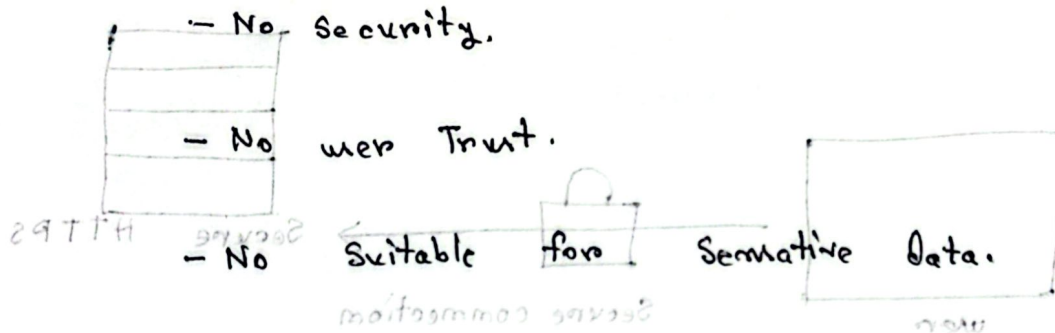
Speed: Faster in small setups, because no encryption

Encryption or decryption is performed.

Compatibility: Supported by all browsers and

Servers, application without extra configuration.

Disadvantages of HTTPS:



HTTPS:

An Extension of HTTP security through Encryption. Anyone can send it. But only the person with right key can open it.

- Port 443 by Default.

- Use Secure Socket Layer or Transport Layer security to encrypt the data.

- Even if attackers capture the message, they cannot read it.

- Browsers show a padlock symbol

next to the URL to indicate that the site is secure.

Advantages:

⇒ Data encryption:

⇒ User trusts

SEO Benefits,

Prevents data tampering.

Support modern protocols.

Disadvantages of HTTPS:

Cost of certificates.

Performance overhead.

Setup complexity - requires configuration of certificates, renewal and proper Server Setup.

Why HTTPS over HTTP:

Data Security.

User trust.

Protection from Hackers.

Better Search Engine Ranking.

Compliance with Standards.

Improved performance.